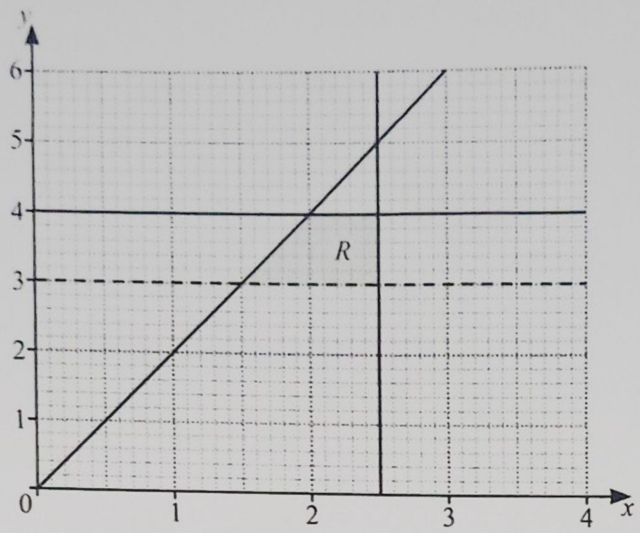
16



Write down all the inequalities that define the region R.

$4 \geq R > 3$

$R \leq 2.5$

$R \leq 4$

*

[4]

17 $I = M(k^2 + c^2)$

(a) Find the value of I when $M = 7$, $k = 3$ and $c = 2$.

$I = 7(3^2 + 2^2)$ $I = 91$

$I = 7(13)$

$I =$ 91 [2]

(b) Rearrange the formula to write k in terms of I, M and c.

$\frac{I}{M} = k^2 + c^2$

$\frac{I}{M} - c^2 = k^2$

$\pm\sqrt{\frac{I}{M} - c^2} = k$

$k = \pm\sqrt{\frac{I}{M} - c^2}$ [3]